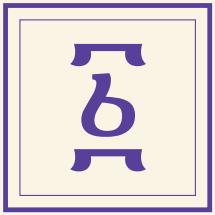
Finding the way, step by step

Weekend 1: Large language models

Carlos Cotrini
AI in Industry, ETH Zurich
Weekend 1, October 2026



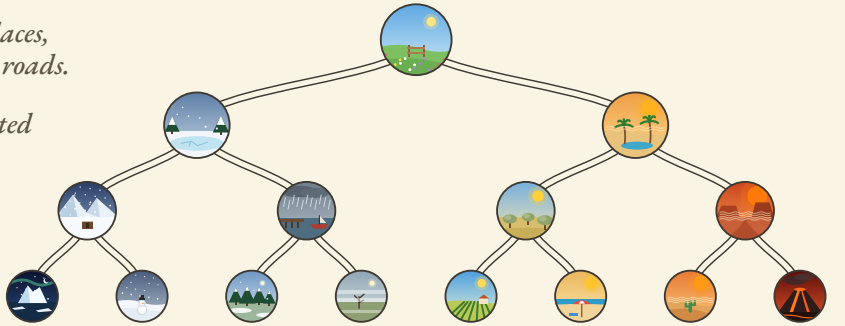
Section 1

The village

Mara carries letters down the valley

Fifteen places, joined by roads.

An invented village.

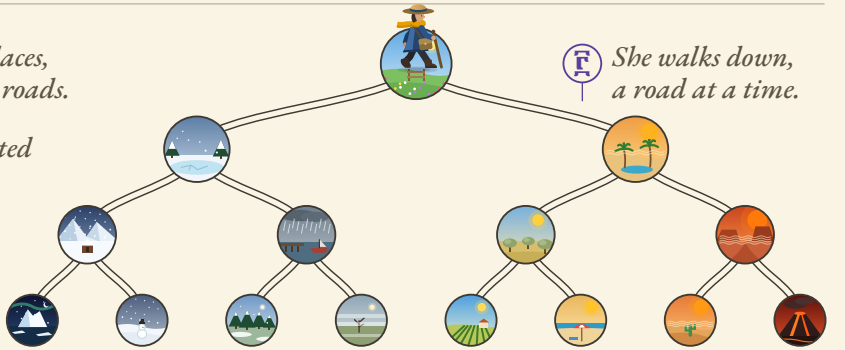


Mara carries letters down the valley

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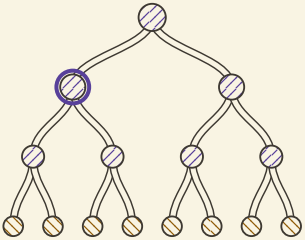
She walks down, a road at a time.



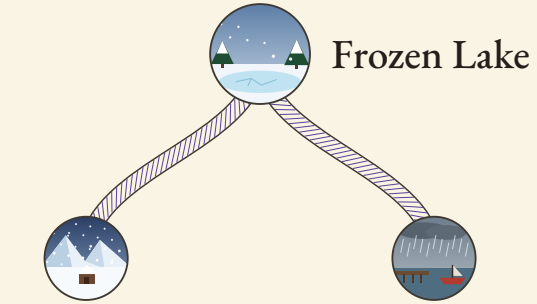
Every place on the way has two roads

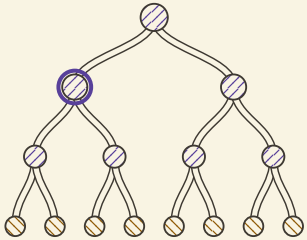


Frozen Lake

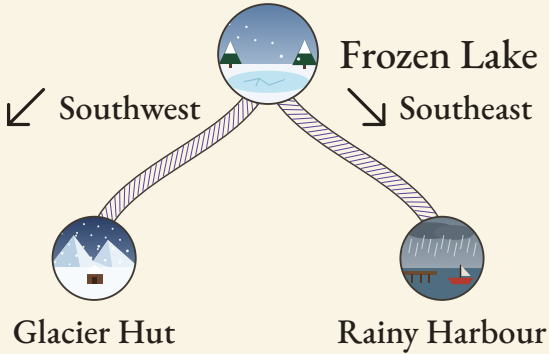


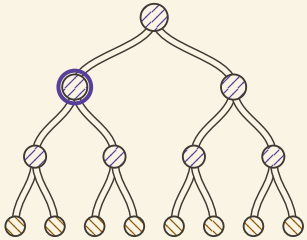
Every place on the way has two roads





Every place on the way has two roads

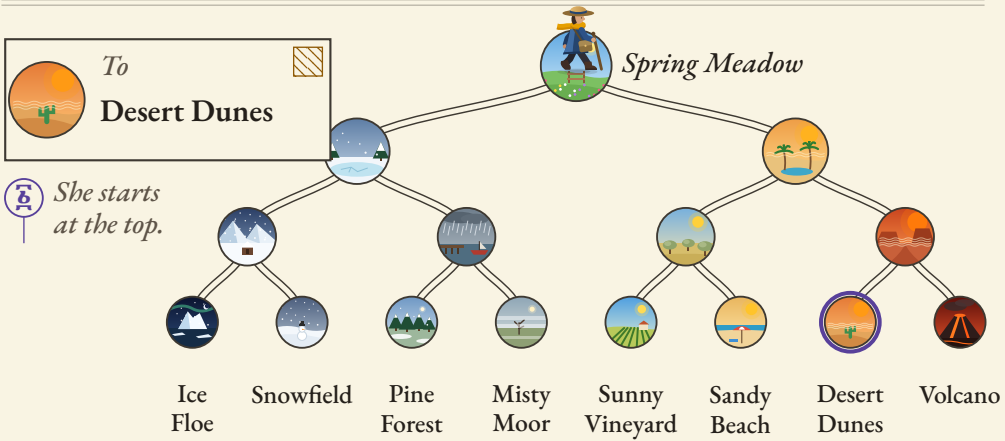




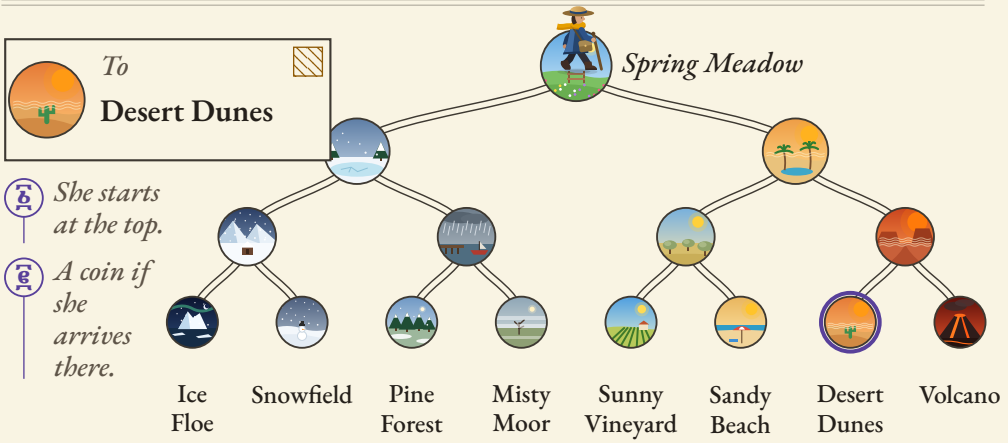
Each letter names a place in the bottom row



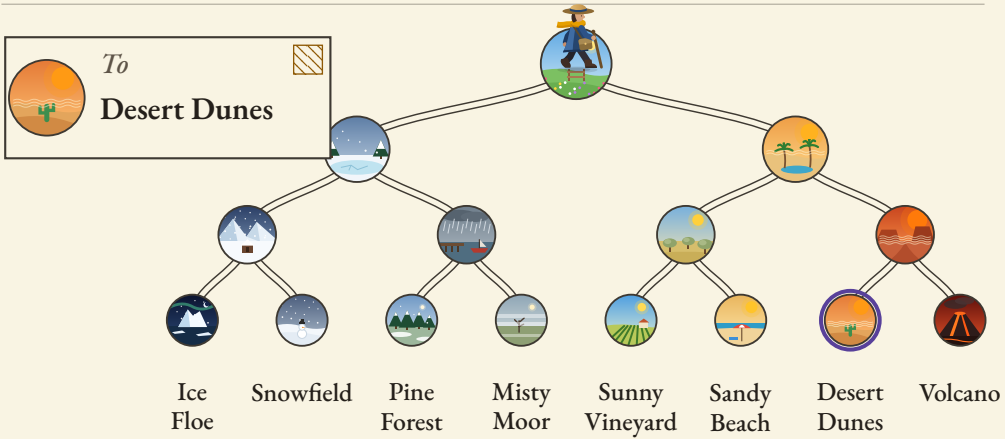
Each letter names a place in the bottom row



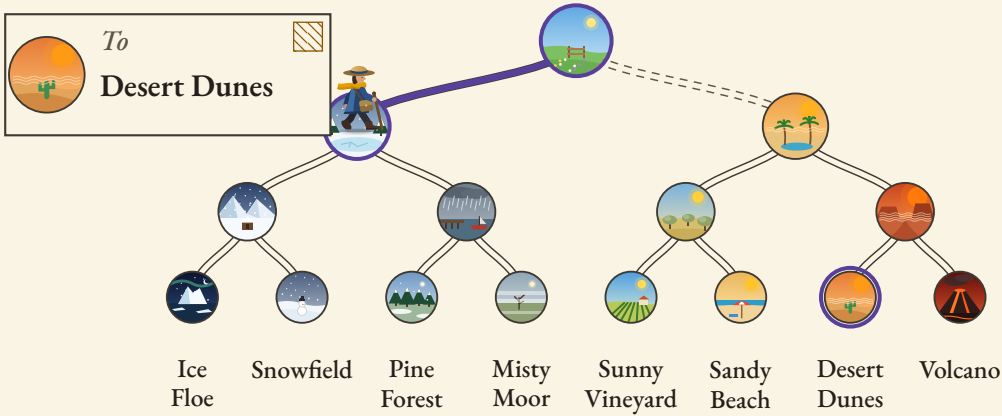
Each letter names a place in the bottom row



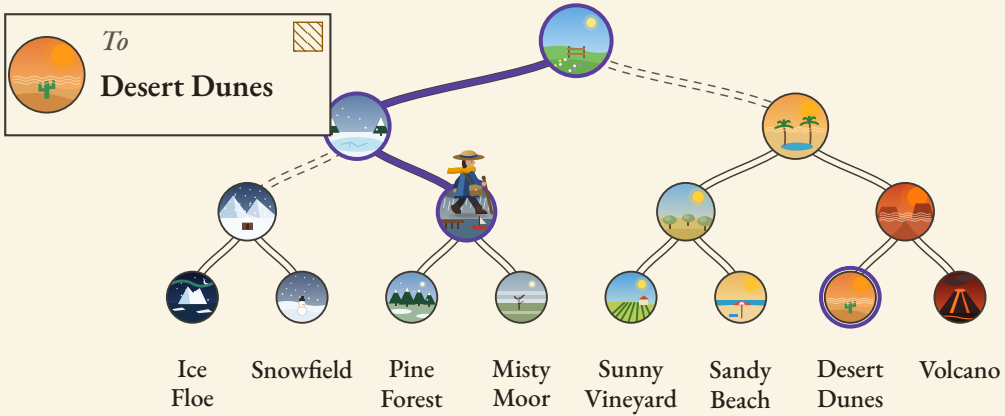
At first she can only guess



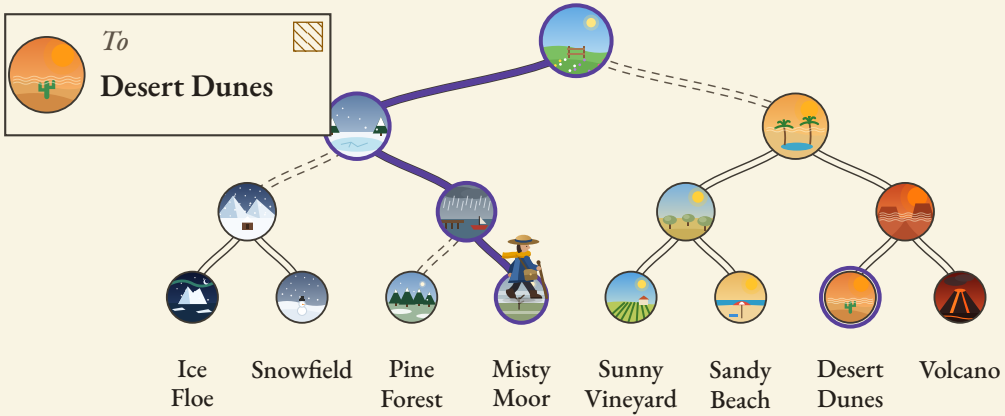
At first she can only guess



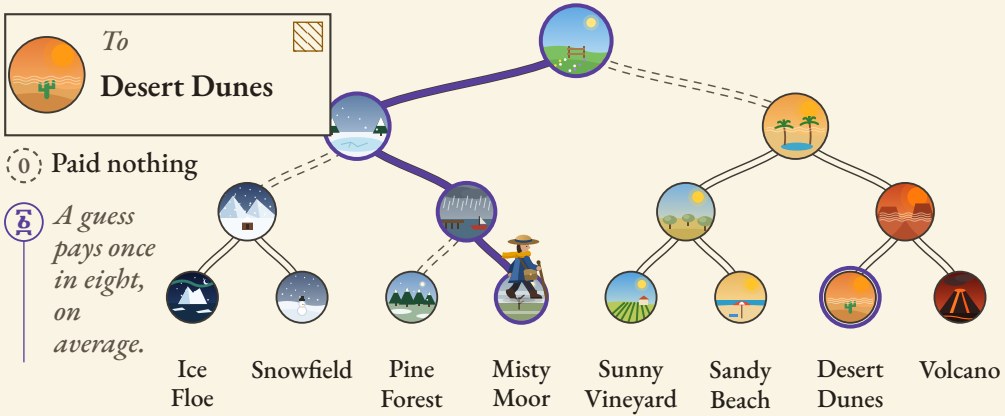
At first she can only guess



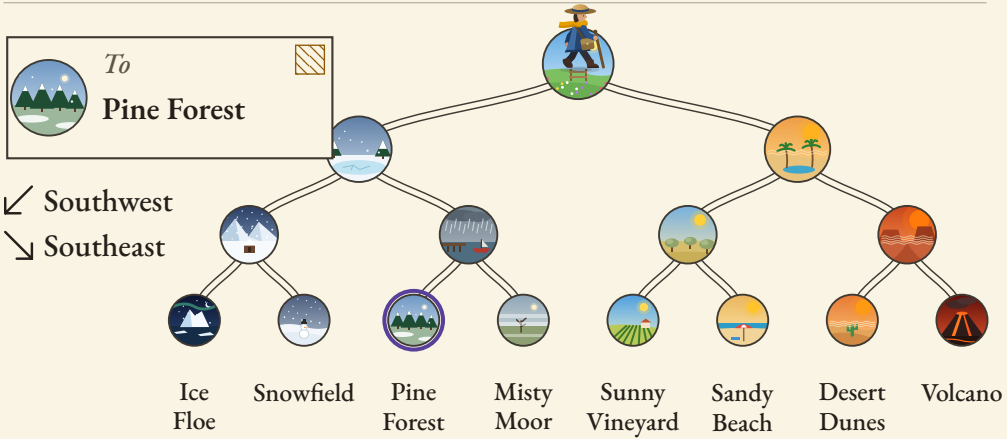
At first she can only guess

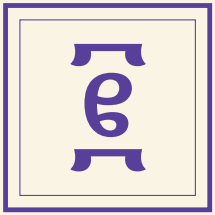


At first she can only guess



Your turn: which road first?





Section 2

The clue

After many letters, a pattern

Letter to	Roads			
Ice Floe	↙	↙	↙	
Snowfield	↙	↙	↘	
Pine Forest	↙	↘	↙	
Misty Moor	↙	↘	↘	
Sunny Vineyard	↘	↙	↙	
Sandy Beach	↘	↙	↘	
Desert Dunes	↘	↘	↙	
Volcano	↘	↘	↘	

She starts at



Spring Meadow

After many letters, a pattern

Letter to	Roads			
Ice Floe	↙	↙	↙	
Snowfield	↙	↙	↘	
Pine Forest	↙	↘	↙	
Misty Moor	↙	↘	↘	
Sunny Vineyard	↘	↙	↙	
Sandy Beach	↘	↙	↘	
Desert Dunes	↘	↘	↙	
Volcano	↘	↘	↘	

She starts at



Spring Meadow

Colder than the start: southwest first.

After many letters, a pattern

Letter to	Roads			
Ice Floe	↙	↙	↙	
Snowfield	↙	↙	↘	
Pine Forest	↙	↘	↙	
Misty Moor	↙	↘	↘	
Sunny Vineyard	↘	↙	↙	
Sandy Beach	↘	↙	↘	
Desert Dunes	↘	↘	↙	
Volcano	↘	↘	↘	

She starts at

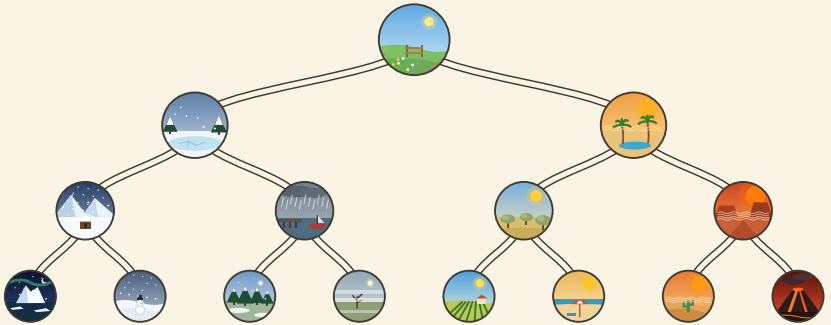


Spring Meadow

Colder than the start: southwest first.

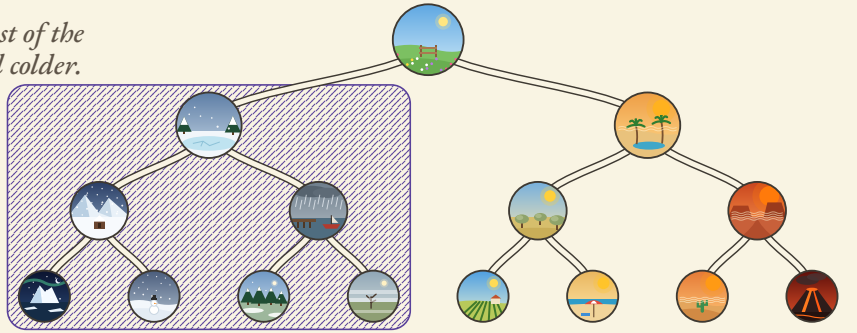
Warmer than the start: southeast first.

Southwest feels colder, southeast feels warmer



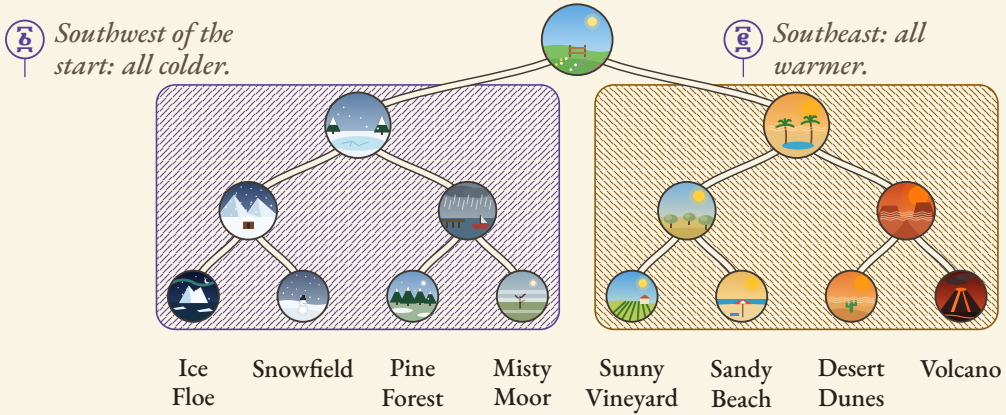
Southwest feels colder, southeast feels warmer

Southwest of the start: all colder.



Ice Floe | Snowfield | Pine Forest | Misty Moor | Sunny Vineyard | Sandy Beach | Desert Dunes | Volcano

Southwest feels colder, southeast feels warmer



The rule: compare here with the letter

Here



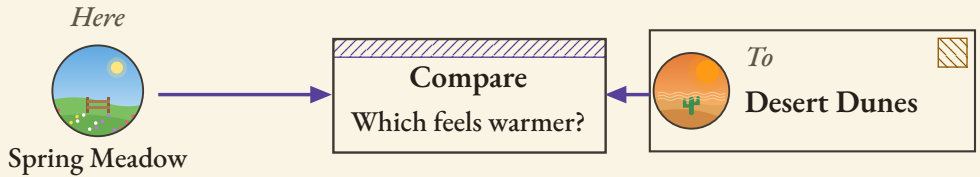
Spring Meadow

To

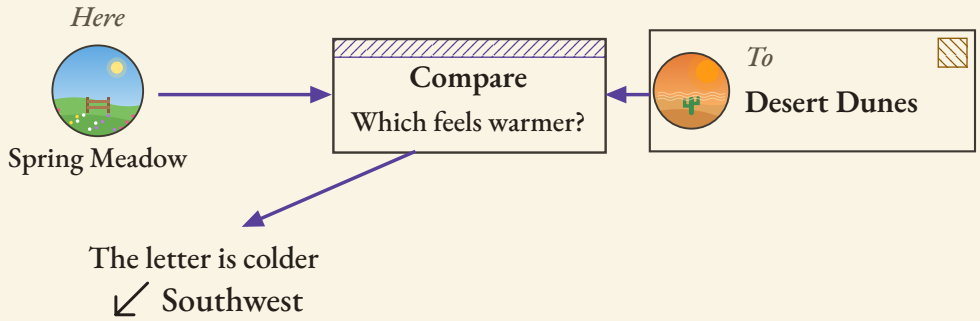


Desert Dunes

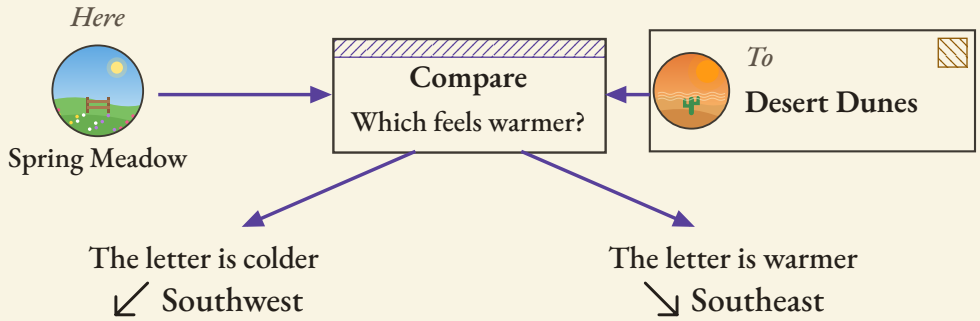
The rule: compare here with the letter



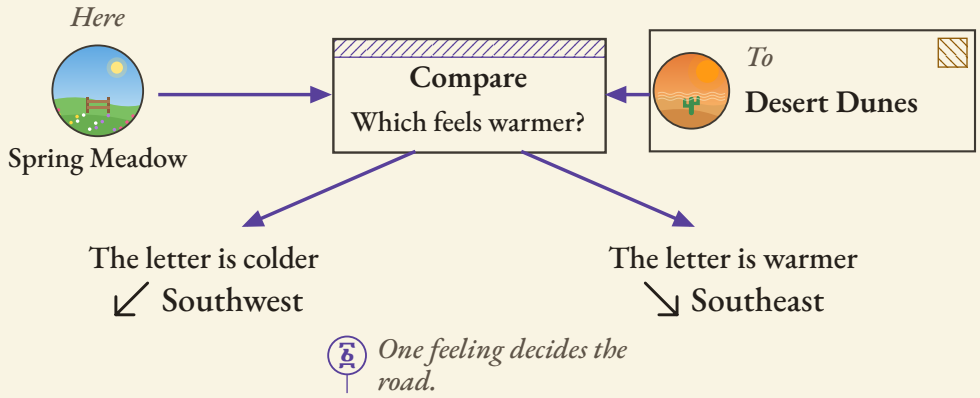
The rule: compare here with the letter



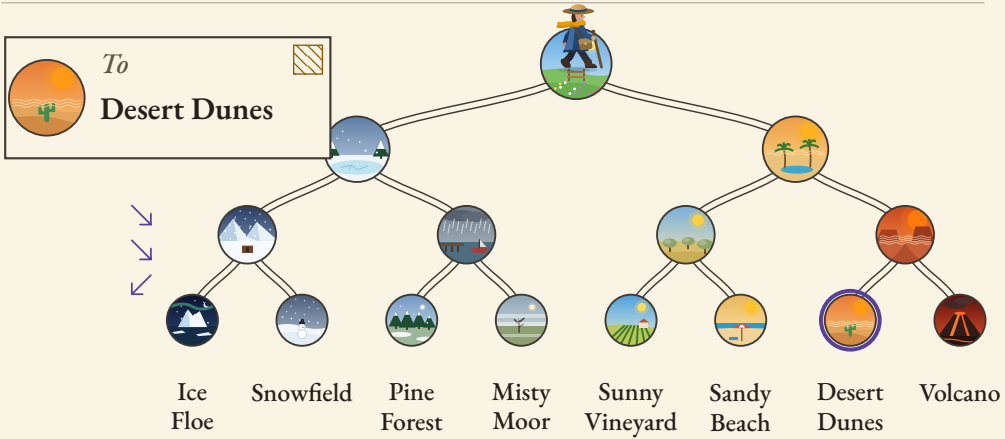
The rule: compare here with the letter



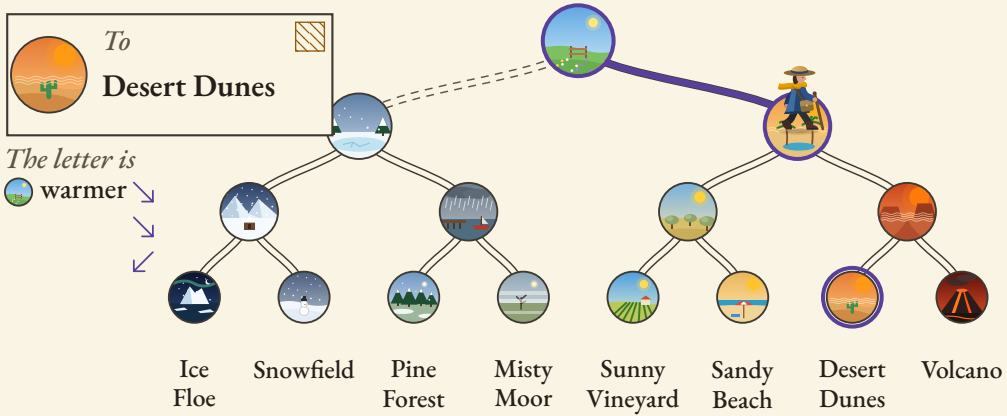
The rule: compare here with the letter



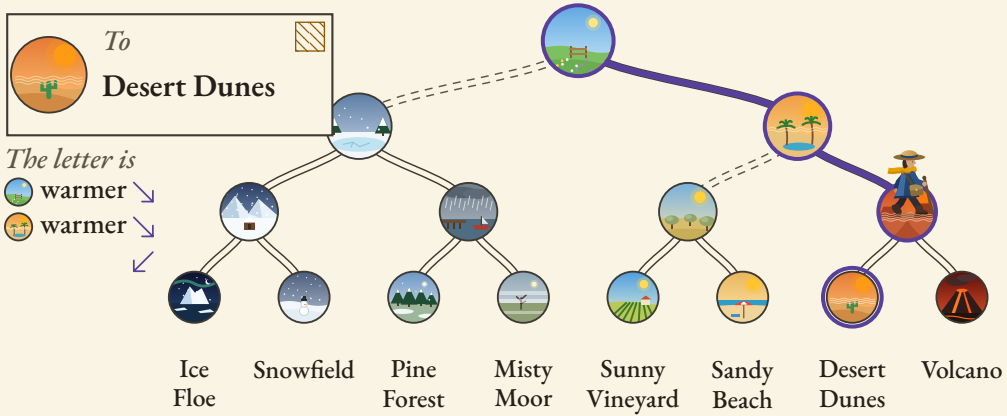
The rule finds Desert Dunes



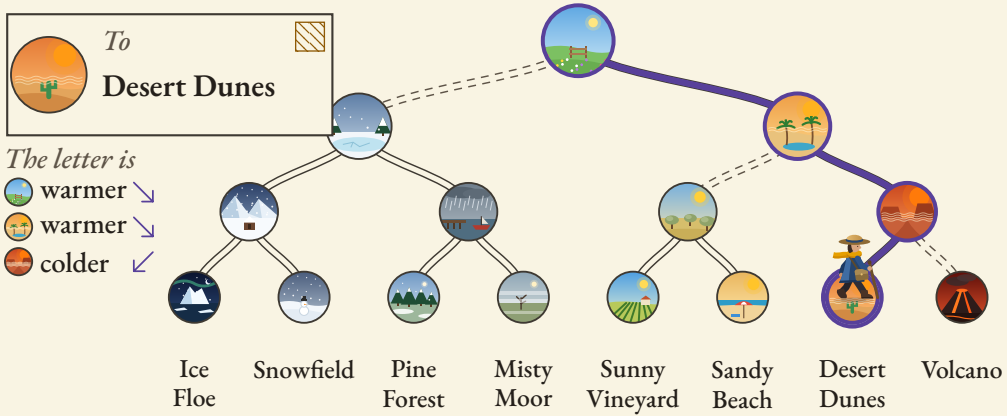
The rule finds Desert Dunes



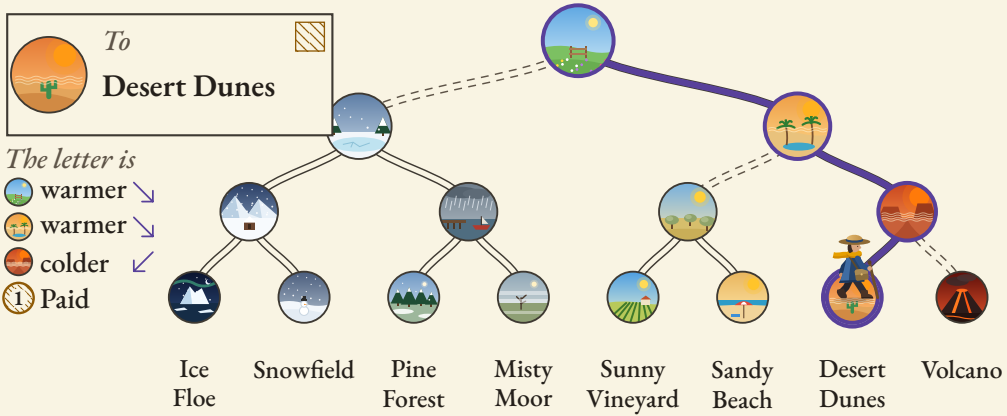
The rule finds Desert Dunes



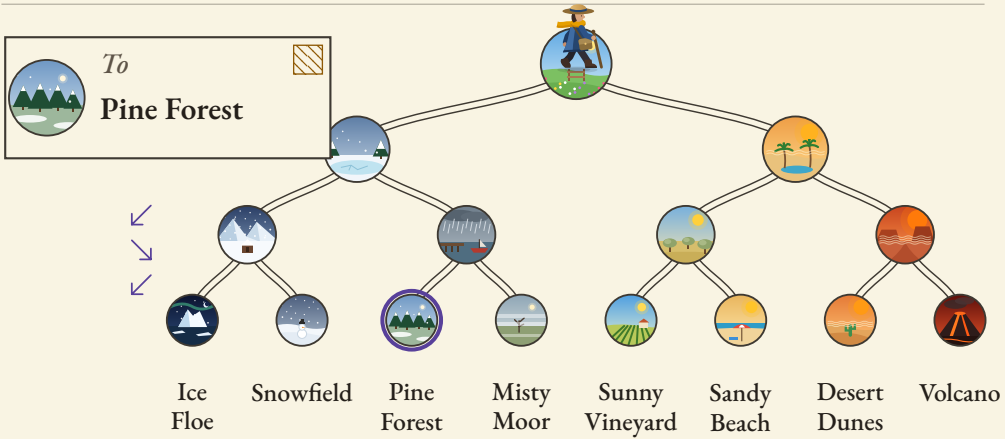
The rule finds Desert Dunes



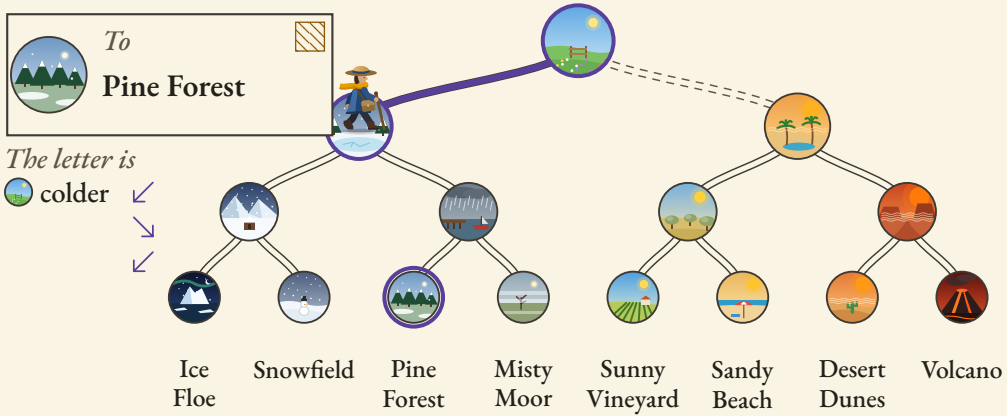
The rule finds Desert Dunes



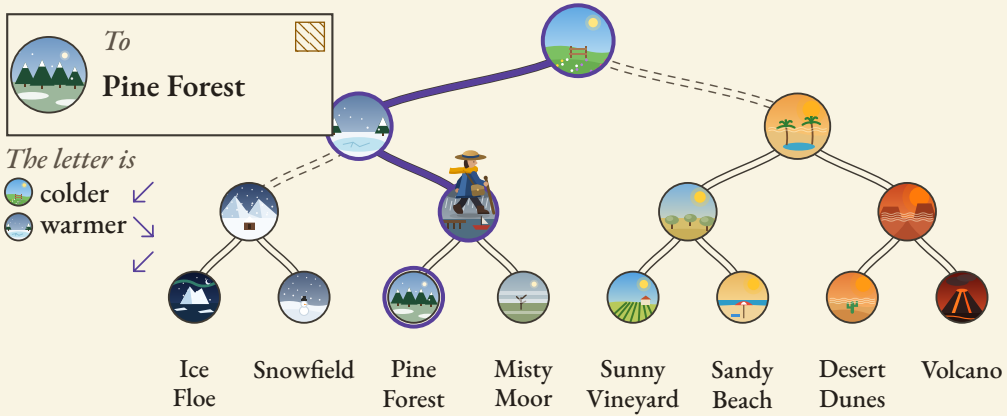
Your turn again: Pine Forest



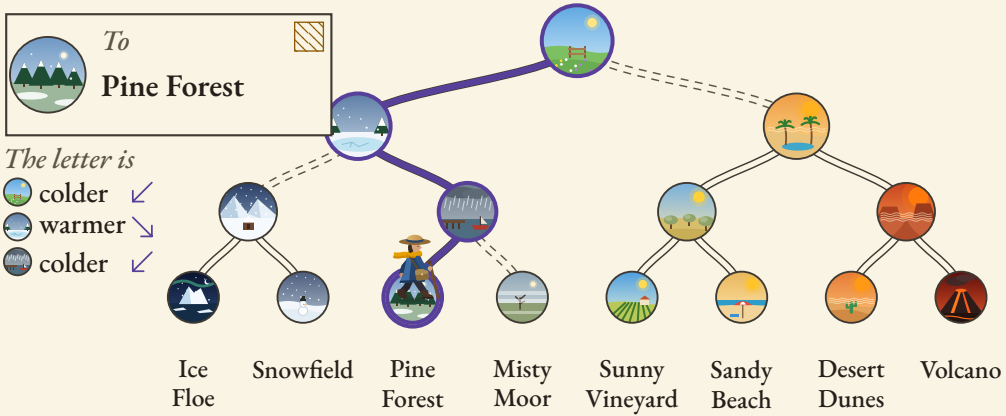
Your turn again: Pine Forest



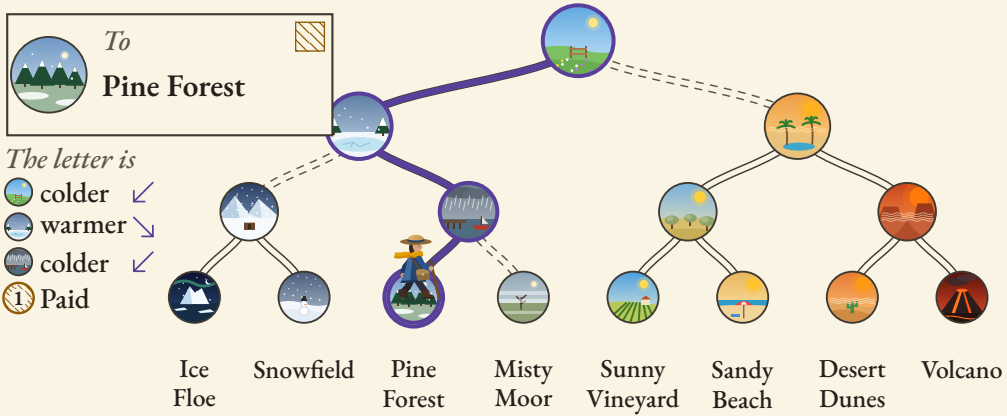
Your turn again: Pine Forest



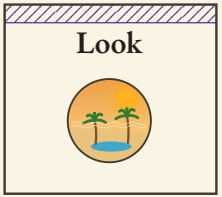
Your turn again: Pine Forest



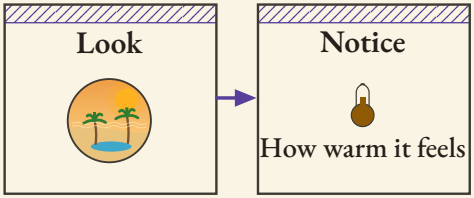
Your turn again: Pine Forest



Look, notice, compare, step

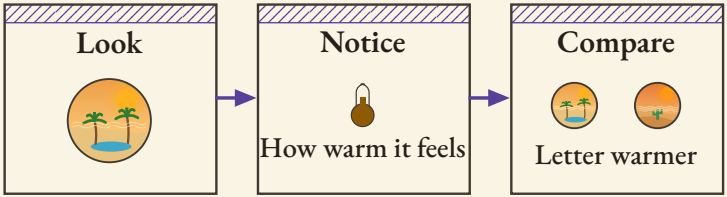


Look, notice, compare, step



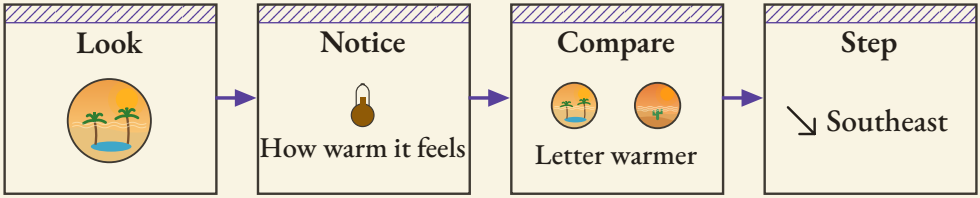
She ignores the rest of the picture.

Look, notice, compare, step



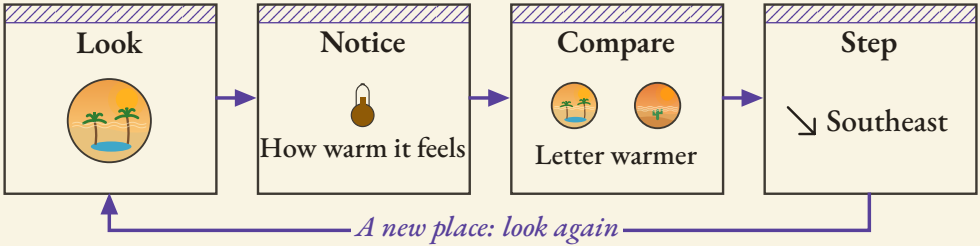
She ignores the rest of the picture.

Look, notice, compare, step



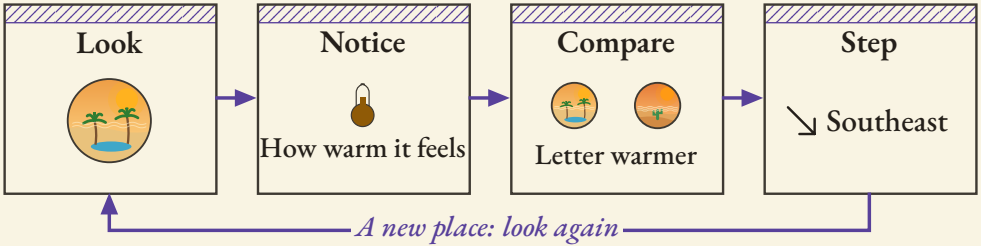
She ignores the rest of the picture.

Look, notice, compare, step



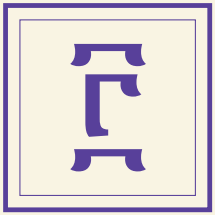
She ignores the rest of the picture.

Look, notice, compare, step



She ignores the rest of the picture.

Nobody told her what to notice. The coins taught her.



Section 3

Reasoning

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	How warm it feels	What matters in the notes

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	How warm it feels	What matters in the notes
Step	A road	One small step, written down

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	How warm it feels	What matters in the notes
Step	A road	One small step, written down
Repeat	Until she arrives	Until it has an answer

A reasoning model runs the same loop

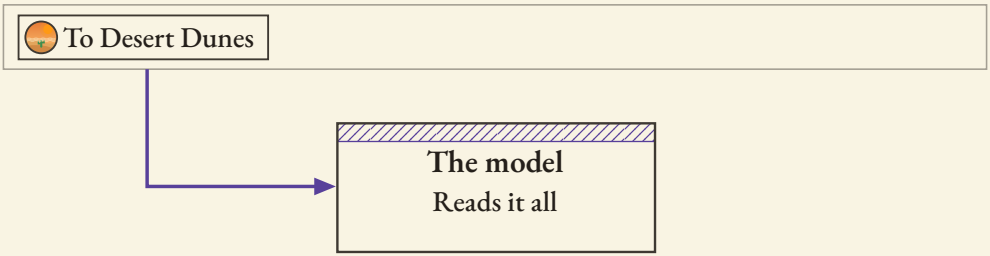
	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	How warm it feels	What matters in the notes
Step	A road	One small step, written down
Repeat	Until she arrives	Until it has an answer
Learn	From the coins	From checking the answers

The model writes one token at a time

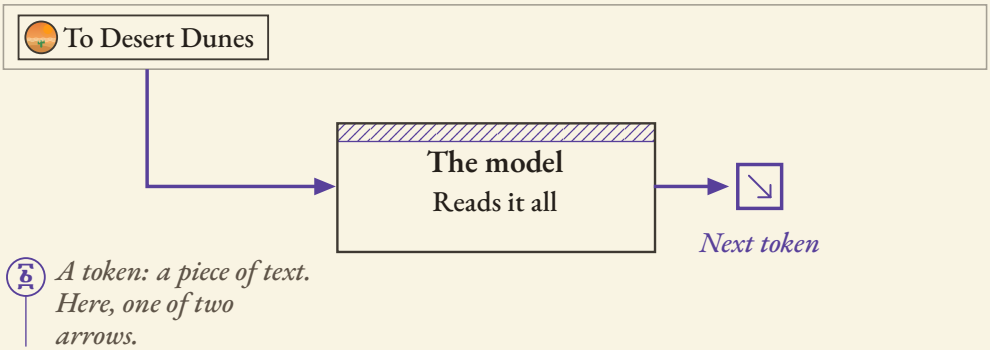
The conversation



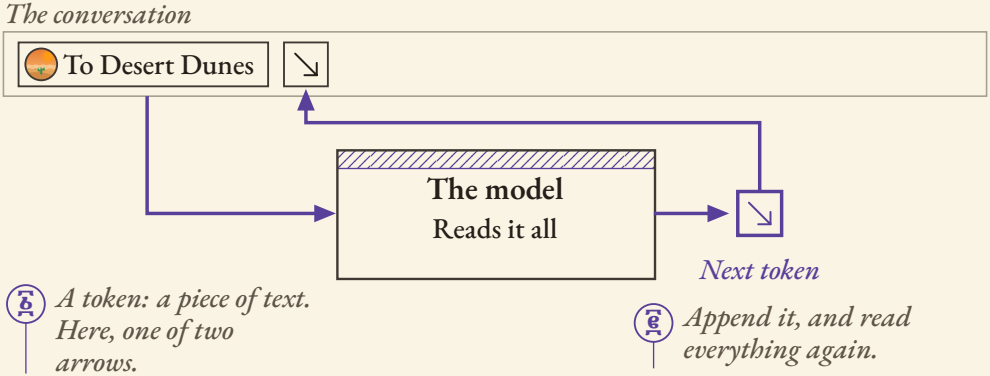
The model writes one token at a time



The model writes one token at a time



The model writes one token at a time



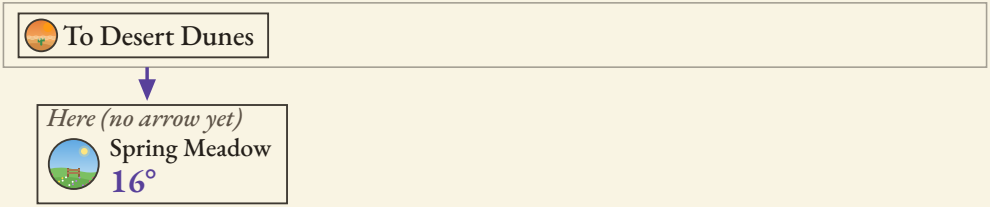
From the conversation, two numbers

The conversation



From the conversation, two numbers

The conversation

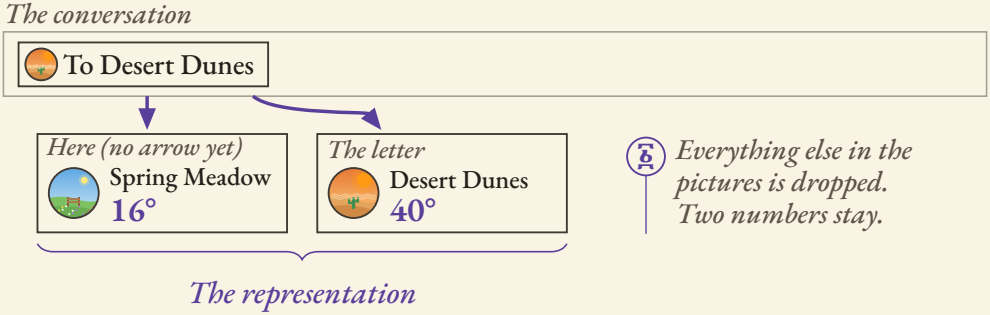


From the conversation, two numbers

The conversation



From the conversation, two numbers



A difference picks the next token

Here



Spring Meadow

16°

The letter



Desert Dunes

40°

A difference picks the next token

Here



Spring Meadow
16°

The letter



Desert Dunes
40°

Letter minus here

$$40 - 16 = +24$$

A difference picks the next token

Here



Spring Meadow
16°

The letter



Desert Dunes
40°

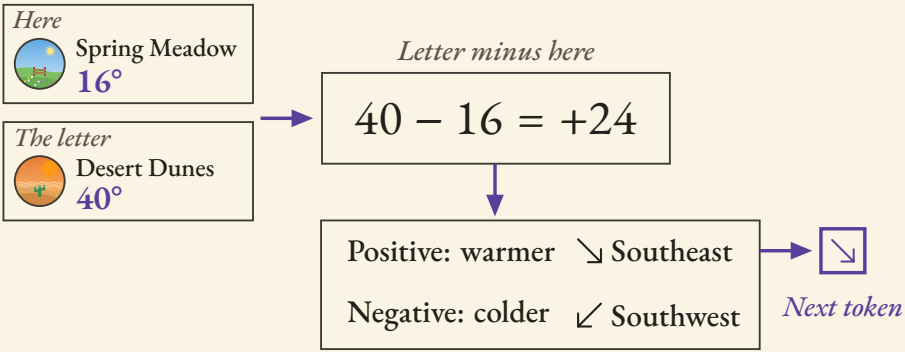
Letter minus here

40 − 16 = +24

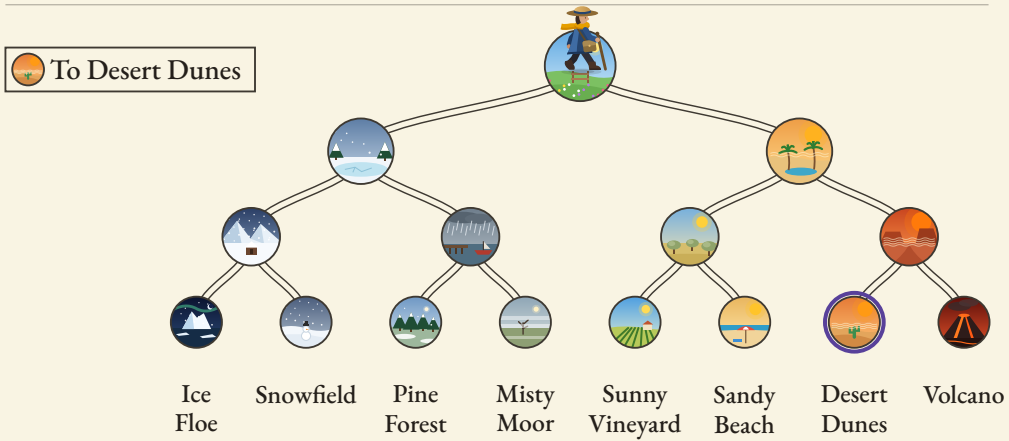
Positive: warmer ↘ Southeast

Negative: colder ↙ Southwest

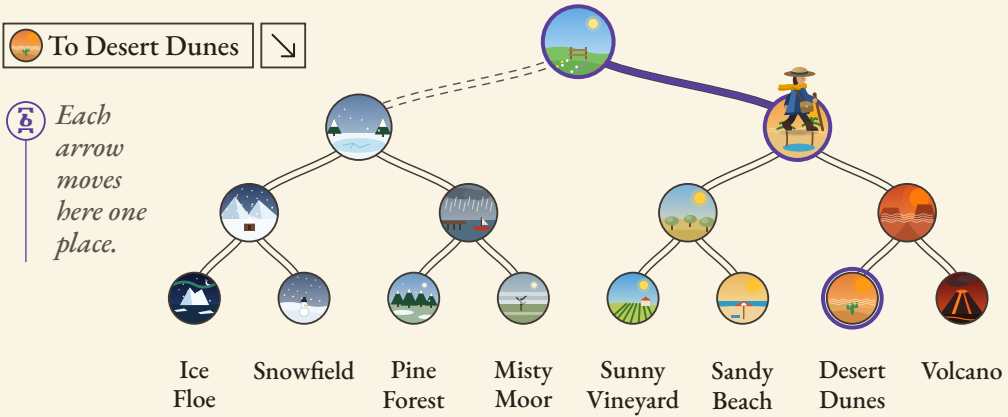
A difference picks the next token



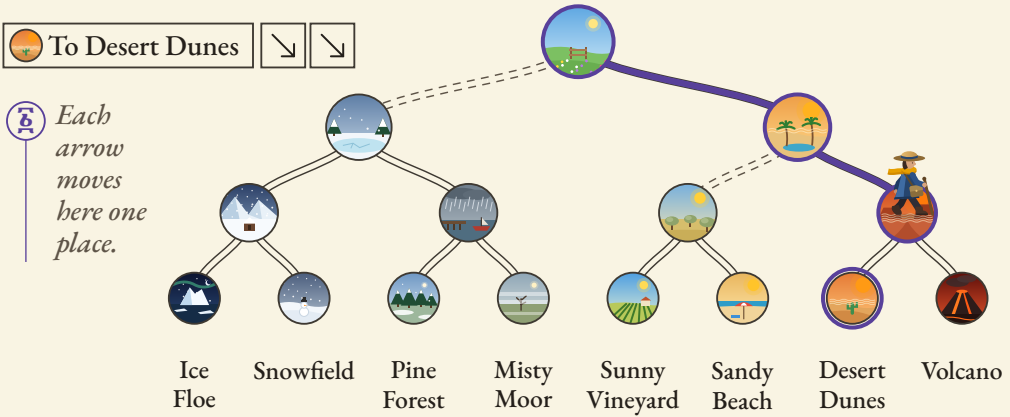
Here is worked out from the tokens



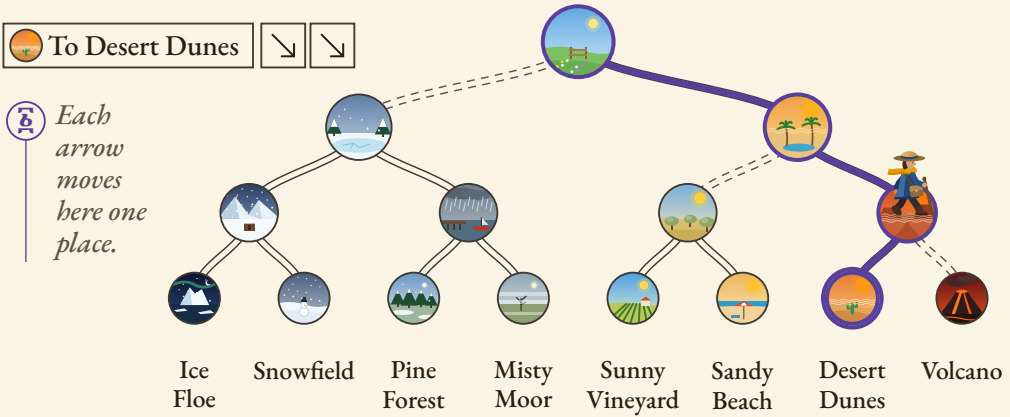
Here is worked out from the tokens



Here is worked out from the tokens



Here is worked out from the tokens



Append, and compute again

The conversation	Here	The letter	Difference	Next
 To Desert Dunes	 16°	 40°	+24	

Append, and compute again

The conversation	Here	The letter	Difference	Next
To Desert Dunes	16°	40°	+24	↘
To Desert Dunes ↘	34°	40°	+6	↘

Each round reads the whole conversation again.

Append, and compute again

The conversation	Here	The letter	Difference	Next
To Desert Dunes	16°	40°	+24	↘
To Desert Dunes ↘	34°	40°	+6	↘
To Desert Dunes ↘ ↘	45°	40°	−5	↙

Each round reads the whole conversation again.

Append, and compute again

The conversation	Here	The letter	Difference	Next
To Desert Dunes	16°	40°	+24	↘
To Desert Dunes ↘	34°	40°	+6	↘
To Desert Dunes ↘ ↘	45°	40°	−5	↙
To Desert Dunes ↘ ↘ ↙	Answer: Desert Dunes			1

Each round reads the whole conversation again.

The arrows before the answer are the thinking.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

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Thinking

3 crates at 40 francs: 120 francs.

Each line reads the notes so far and adds one small step.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.
10 percent of 120 is 12.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.
10 percent of 120 is 12.
120 minus 12 is 108.

Each line reads the notes so far and adds one small step.

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In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

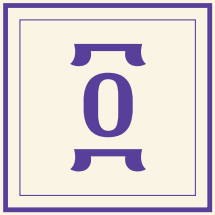
Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.
10 percent of 120 is 12.
120 minus 12 is 108.

Answer: the client pays 108 francs.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.



Section 4

Training

The model keeps a table of chances

Here	The letter	Southwest	Southeast
Spring Meadow	Ice Floe		
Spring Meadow	Desert Dunes		
Frozen Lake	Pine Forest		
Palm Oasis	Sunny Vineyard		
Red Canyon	Desert Dunes		
Rainy Harbour	Pine Forest		

And 18 more rows

A row for each place and letter: twenty-four rows.

The model keeps a table of chances

Here	The letter	Southwest	Southeast
Spring Meadow	Ice Floe	21%	79%
Spring Meadow	Desert Dunes	67%	33%
Frozen Lake	Pine Forest	14%	86%
Palm Oasis	Sunny Vineyard	75%	25%
Red Canyon	Desert Dunes	86%	14%
Rainy Harbour	Pine Forest	81%	19%

And 18 more rows



A row for each place and letter: twenty-four rows.

At first, random.

One iteration: pick a row at random

Here	The letter	Southwest	Southeast
 Spring Meadow	 Ice Floe	21%	79%
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Rainy Harbour	Pine Forest	81%	19%

And 18 more rows



Picked at random.

Step two: send walks, half each way

Here



Palm Oasis



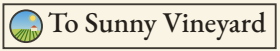
To Sunny Vineyard

Step two: send walks, half each way

Here

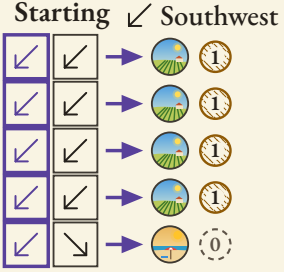


Palm Oasis





Later roads: drawn from the table.



Step two: send walks, half each way

Here



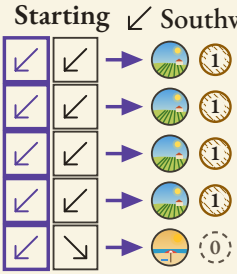
Palm Oasis

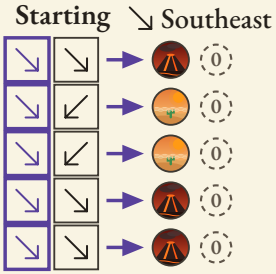


To Sunny Vineyard



Later roads: drawn from the table.





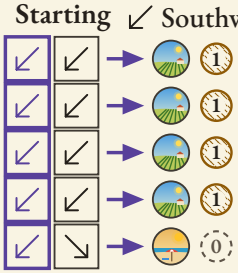
Step two: send walks, half each way

Here

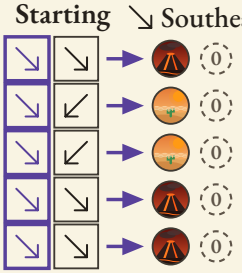
Palm Oasis

To Sunny Vineyard

Later roads: drawn from the table.

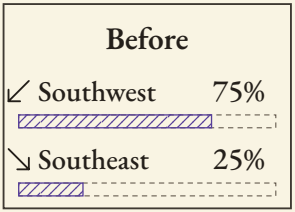


4 of 5 arrived

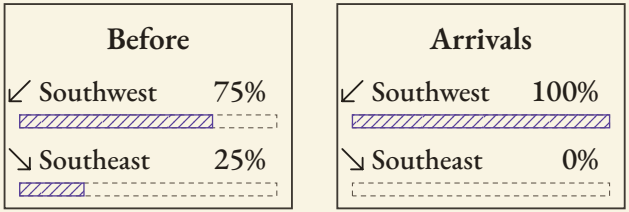


0 of 5 arrived

Step three: move the row toward what arrived

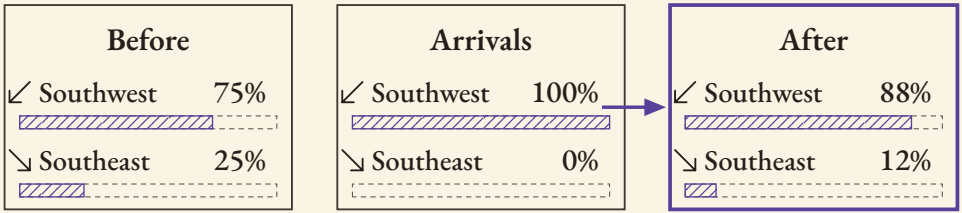


Step three: move the row toward what arrived



4 walks arrived

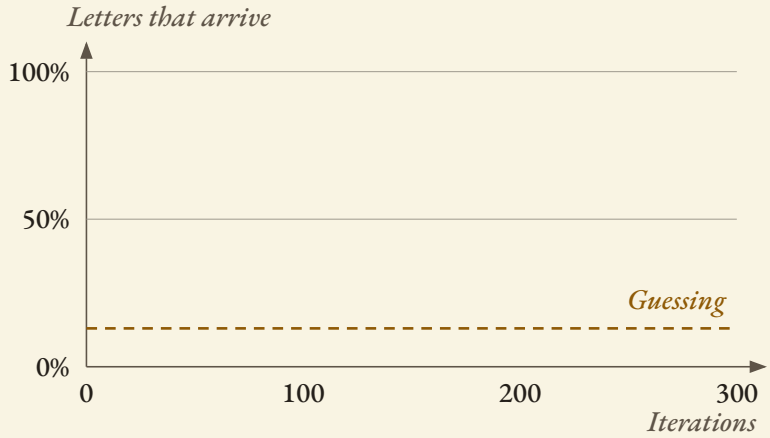
Step three: move the row toward what arrived



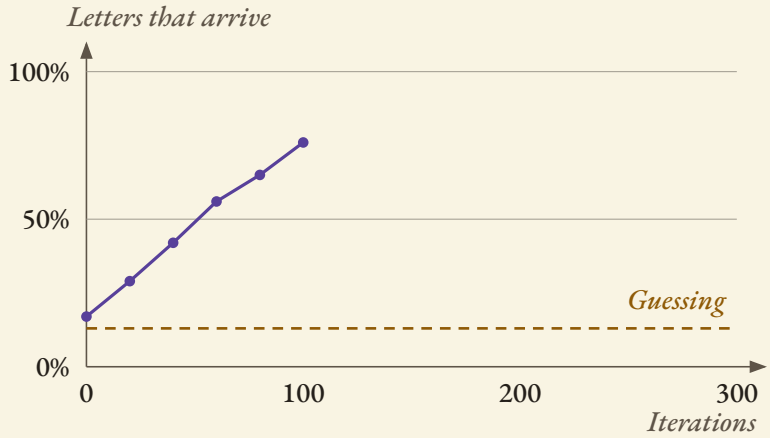
4 walks arrived

After: halfway from before to the arrivals.

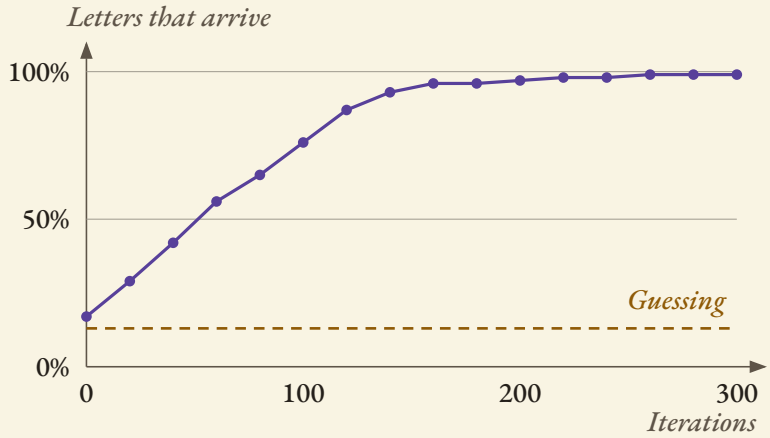
Repeat, many times



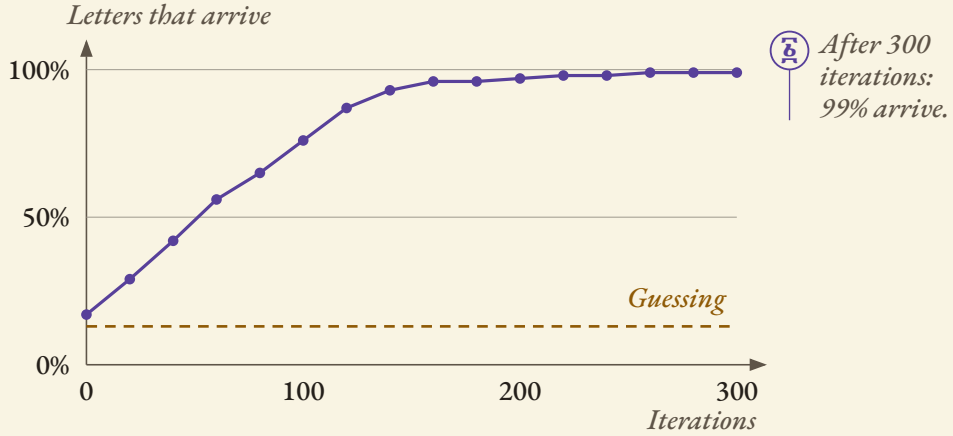
Repeat, many times



Repeat, many times



Repeat, many times



After training, the table knows the rule

Here	The letter	Southwest	Southeast
 Spring Meadow	 Ice Floe		
 Spring Meadow	 Desert Dunes		
 Frozen Lake	 Pine Forest		
 Palm Oasis	 Sunny Vineyard		
 Red Canyon	 Desert Dunes		
 Rainy Harbour	 Pine Forest		

And 18 more rows

After training, the table knows the rule

Here	The letter	Southwest	Southeast
Spring Meadow	Ice Floe	100%	0%
Spring Meadow	Desert Dunes	1%	99%
Frozen Lake	Pine Forest	1%	99%
Palm Oasis	Sunny Vineyard	100%	0%
Red Canyon	Desert Dunes	100%	0%
Rainy Harbour	Pine Forest	100%	0%

And 18 more rows

Colder letter: southwest. Warmer: southeast.

Learned from the coins alone.

A check pays the answer

Answers from training

A check pays the answer

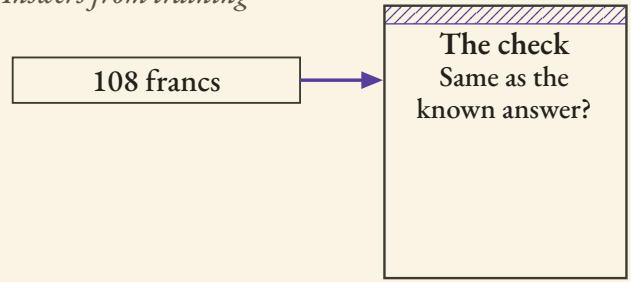
Answers from training

108 francs



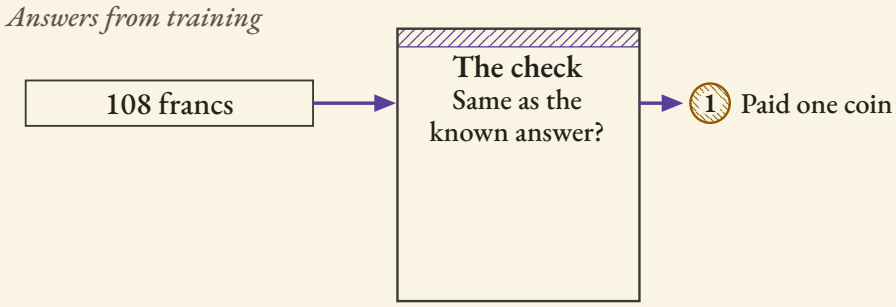
Training uses questions whose answer is known.

A check pays the answer

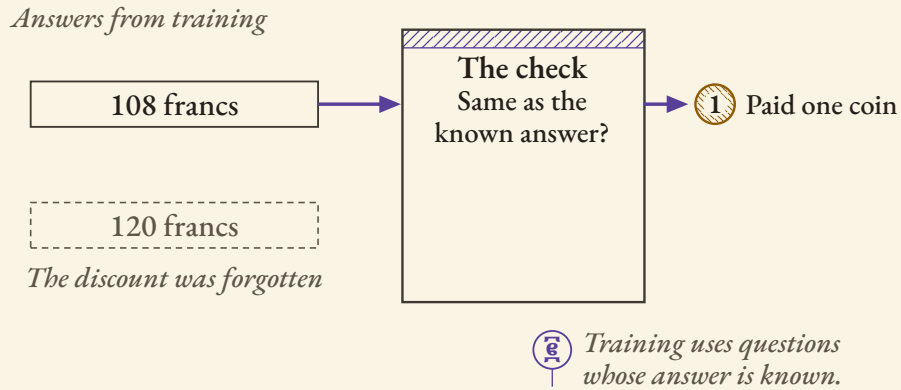


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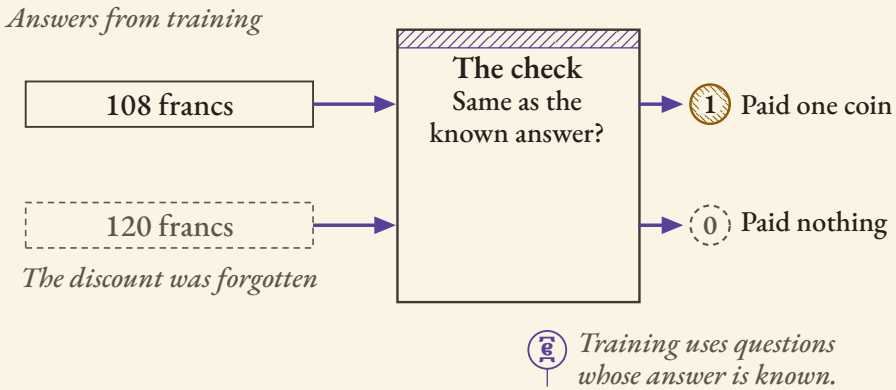
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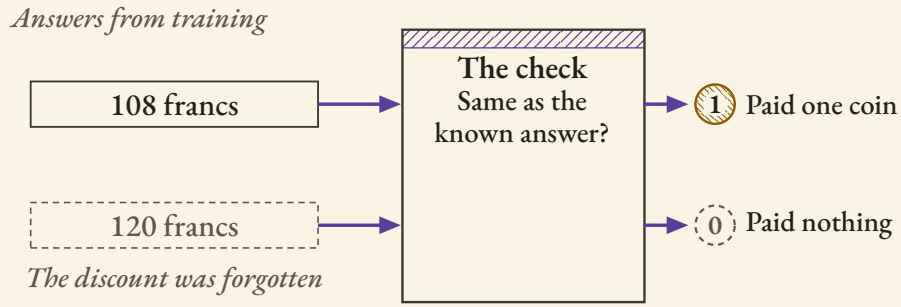
A check pays the answer



A check pays the answer



A check pays the answer



The check reads the answer only. Nobody grades the notes.

Training uses questions whose answer is known.

A model learns its notes from the check

Write many answers

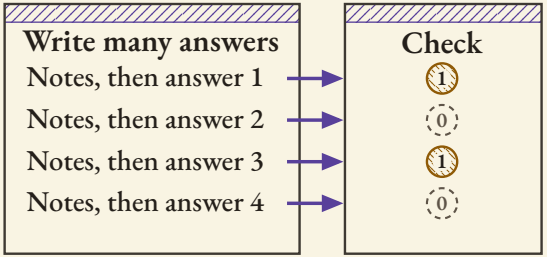
Notes, then answer 1

Notes, then answer 2

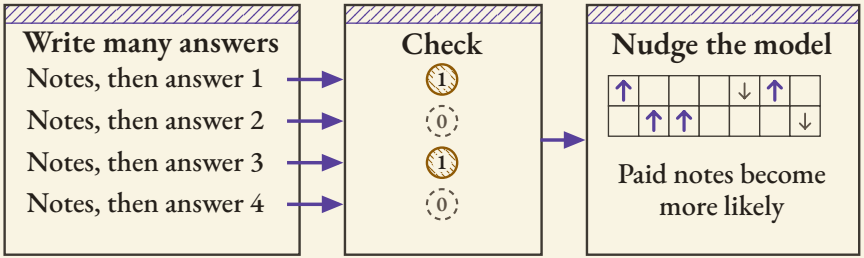
Notes, then answer 3

Notes, then answer 4

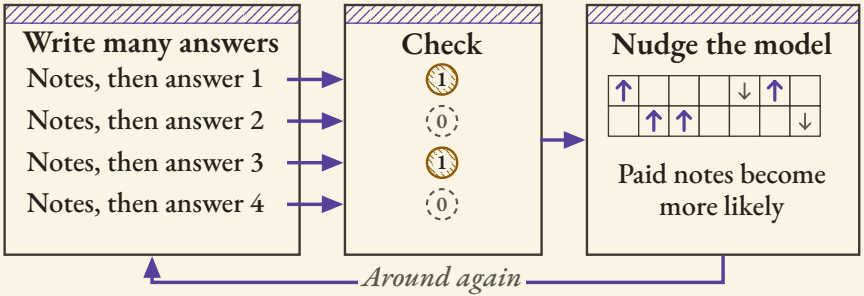
A model learns its notes from the check



A model learns its notes from the check



A model learns its notes from the check



Mara learned the clue from her coins.

A model learns its notes from the check.

DeepSeek-R1 learned to reason from a check

https://doi.org/10.1038/s41586-025-09422-z

D. Guo et al. (DeepSeek-AI), 2025. *DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning*. *Nature* 645, 633-638.

DeepSeek-R1 learned to reason from a check

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- Reasoning appeared by itself: longer notes, checking its own work, trying another way.
- The released DeepSeek-R1 adds some written examples and more training.

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The idea behind most reasoning models

- Look at where you stand

Example: D. Guo et al., *Nature* 645, 633-638 (2025), doi:10.1038/s41586-025-09422-z.

The idea behind most reasoning models

- Look at where you stand
- Notice what matters

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- Look at where you stand
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- Take one small step, and write it down

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- Look at where you stand
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- Repeat until there is an answer

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The idea behind most reasoning models

- Look at where you stand
- Notice what matters
- Take one small step, and write it down
- Repeat until there is an answer
- Learn from a check of the answer which steps pay off

Example: D. Guo et al., *Nature* 645, 633-638 (2025), doi:10.1038/s41586-025-09422-z.